

Adrian M. Price-Whelan — Publication List

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Publications — ADS search

Refereed: 164 articles (18 first author)

Citations 58547

h-index: 59

(as of 2026-09-07)

Refereed

- 164 Arora, A. *et al.* (incl. **APW**), *No Stream Left Unscathed: The Imprint of a Host Galaxy*, ApJ, 1008, 91, 2026 ([arXiv:2605.16200](#))
- 163 Holm, J. K. *et al.* (incl. **APW**), *Evidence for the first globular cluster stellar stream beyond the Milky Way*, Nature, 656, 843, 2026
- 162 Rose, J. C. *et al.* (incl. **APW**), *The DREAMS Project: Disentangling the Impact of Halo-to-halo Variance and Baryonic Feedback on Milky Way Satellite Galaxies*, ApJ, 1006, 244, 2026 ([arXiv:2512.02095](#)) [3 citations]
- 161 Brooks, R. A. N. *et al.* (incl. **APW**), *The Milky Way–Large Magellanic Cloud interaction with simulation-based inference*, MNRAS, 550, 2026 ([arXiv:2510.04735](#)) [9 citations]
- 160 Brooks, R. A. N. *et al.* (incl. **APW**), *Quantifying the Milky Way, LMC and their interaction using all-sky kinematics of outer halo stars*, MNRAS, 549, 2026 ([arXiv:2603.08788](#)) [5 citations]
- 159 SDSS Collaboration *et al.* (incl. **APW**), *The Nineteenth Data Release of the Sloan Digital Sky Survey*, ApJS, 285, 9, 2026 ([arXiv:2507.07093](#)) [80 citations]
- 158 Sagar, S.; Ballard, S.; Daniel, K. J.; **Price-Whelan, A. M.** *et al.*, *The Orbital Eccentricities of Planets in the Kinematic Thin and Thick Galactic Disks*, AJ, 172, 42, 2026
- 157 Nibauer, J.; Bonaca, A.; **Price-Whelan, A. M.**; Spergel, D. N. *et al.*, *Measurement of Substructure from the Kinematics of the GD-1 Stellar Stream*, ApJ, 1004, 62, 2026 ([arXiv:2510.02247](#)) [20 citations]
- 156 Chen, Y.; Gnedin, O. Y.; **Price-Whelan, A. M.**, *StarStream on Gaia: Stream Discovery and Mass-loss Rate of Globular Clusters*, ApJS, 283, 60, 2026 ([arXiv:2510.14924](#)) [7 citations]
- 155 Lucey, M.; Mateu, C.; **Price-Whelan, A. M.**; Hogg, D. W. *et al.*, *Inferring Stellar Densities with Flexible Models. I. The Distribution of RR Lyrae in the Milky Way with Gaia DR3*, AJ, 171, 249, 2026 ([arXiv:2510.03221](#)) [4 citations]
- 154 Horta, D.; **Price-Whelan, A. M.**; Koposov, S. E.; Hunt, J. A. S. *et al.*, *The Milky Way's Circular Velocity Curve Measured Using Element Abundance Gradients*, ApJ, 1000, 8, 2026 ([arXiv:2601.18876](#)) [2 citations]

- 153 Hughes, J. et al. (incl. **APW**), *The Milky Way Bulge Extratidal Star Survey: NGC 6569*, AJ, 171, 137, 2026 (arXiv:2512.19074) [2 citations]
- 152 Otto, J. M. et al. (incl. **APW**), *The Open Cluster Chemical Abundances and Mapping Survey. VIII. Galactic Chemical Gradient and Azimuthal Analysis from SDSS/MWM DR19*, AJ, 171, 91, 2026 (arXiv:2507.07264) [22 citations]
- 151 Brooks, R. A. N. et al. (incl. **APW**), *A simulation-based inference approach to the dynamics of the MW-LMC system - validation*, MNRAS, 545, 2026 (arXiv:2507.10667) [15 citations]
- 150 Kollmeier, J. A. et al. (incl. **APW**), *Sloan Digital Sky Survey. V. Pioneering Panoptic Spectroscopy*, AJ, 171, 52, 2026 (arXiv:2507.06989) [107 citations]
- 149 Hunt, J. A. S. et al. (incl. **APW**), *The dark matter wake of a galactic bar revealed by multichannel singular spectral analysis*, MNRAS, 545, 2026 (arXiv:2510.09751) [6 citations]
- 148 Chen, Y.; Gnedin, O. Y.; **Price-Whelan, A. M.**; Holm-Hansen, C., *StarStream: Automatic Detection Algorithm for Stellar Streams*, ApJ, 995, 15, 2025 (arXiv:2510.14929) [5 citations]
- 147 Viswanathan, A.; Horta, D.; **Price-Whelan, A. M.**; Starkenburg, E., *A slow spin to win: The gradual kinematic evolution across metallicities of the proto-Galaxy to the high- α disc*, A&A, 703, 2025 (arXiv:2411.12165) [14 citations]
- 146 Oeur, M.; Loebman, S. R.; **Price-Whelan, A. M.**; Arora, A. et al., *OTI on FIRE: Testing the Efficacy of Orbital Torus Imaging to Recover the Galactic Potential*, ApJ, 993, 29, 2025 (arXiv:2505.05590) [2 citations]
- 145 Bernet, M.; Ramos, P.; Antoja, T.; **Price-Whelan, A. M.** et al., *Dynamics of tidal spiral arms: Machine learning-assisted identification of equations and application to the Milky Way*, A&A, 702, 2025 (arXiv:2506.17383) [6 citations]
- 144 Chance, Q. et al. (incl. **APW**), *paired: A Statistical Framework for Detecting Stellar Binarity with Gaia RVs. I. Sensitivity to Unresolved Binaries*, ApJ, 992, 131, 2025 (arXiv:2206.11275) [12 citations]
- 143 Adams, D. K. et al. (incl. **APW**), *Semianalytic Modeling of Dark Matter Subhalo Encounters with Thin Stellar Streams: Statistical Predictions for GD-1-like Streams in Cold Dark Matter*, ApJ, 991, 66, 2025 (arXiv:2412.13144) [6 citations]
- 142 Sheffield, A. A. et al. (incl. **APW**), *Tracing Red Giant Members of the Globular Cluster Palomar 5 with APOGEE and Gaia*, ApJ, 991, 30, 2025 (arXiv:2507.21212)
- 141 Wagg, T. et al. (incl. **APW**), *Delayed and Displaced: The Impact of Binary Interactions on Core-collapse SN Feedback*, AJ, 170, 192, 2025 (arXiv:2504.17903) [12 citations]
- 140 Anguiano, B. et al. (incl. **APW**), *Tidal Debris Candidates from the ω Centauri Accretion Event and Its Role Building Up the Milky Way Halo*, ApJ, 989, 12, 2025 (arXiv:2505.08353) [4 citations]
- 139 Fillion, C. et al. (incl. **APW**), *Counterculture Stars: Slow and Retrograde Stars with Low- α Disk Abundances*, ApJ, 989, 70, 2025 (arXiv:2506.09927)
- 138 Tavangar, K.; **Price-Whelan, A. M.**, *Inferring the Density and Membership of Stellar Streams with Flexible Models: The GD-1 Stream in Gaia Data Release 3*, ApJ, 988, 45, 2025 (arXiv:2502.13236) [13 citations]
- 137 Frankel, N.; Hogg, D. W.; Tremaine, S.; **Price-Whelan, A. M.** et al., *Iron Snails: Nonequilib-*

- rium Dynamics and Spiral Abundance Patterns*, ApJ, 987, 81, 2025 (arXiv:2407.07149) [18 citations]
- 136 Horta, D.; **Price-Whelan, A. M.**; Hogg, D. W.; Ness, M. K. et al., *Lux: A Generative, Multi-output, Latent-variable Model for Astronomical Data with Noisy Labels*, AJ, 169, 314, 2025 (arXiv:2502.01745) [10 citations]
- 135 Bonaca, A.; **Price-Whelan, A. M.**, *Stellar streams in the Gaia era*, New Astronomy Reviews, 100, 101713, 2025 (arXiv:2405.19410) [95 citations]
- 134 Griffith, E. J. et al. (incl. **APW**), *Many Elements Matter: Detailed Abundance Patterns Reveal Star Formation and Enrichment Differences among Milky Way Structural Components*, AJ, 169, 280, 2025 (arXiv:2410.22121) [16 citations]
- 133 Widrow, L. M. et al. (incl. **APW**), *An Equilibrium Model of the Galaxy Determined by Element Abundance Gradients*, ApJ, 983, 134, 2025
- 132 Nibauer, J.; Bonaca, A.; Spergel, D. N.; **Price-Whelan, A. M.** et al., *StreamSculptor: Hamiltonian Perturbation Theory for Stellar Streams in Flexible Potentials with Differentiable Simulations*, ApJ, 983, 68, 2025 (arXiv:2410.21174) [18 citations]
- 131 Starkman, N.; **Price-Whelan, A. M.**; Nibauer, J., *unxt: A Python package for unit-aware computing with JAX*, JOSS, 10, 7771, 2025 (arXiv:2603.08770)
- 130 Lu, Y. et al. (incl. **APW**), *Evidence of Truly Young High- α Dwarf Stars*, AJ, 169, 168, 2025 (arXiv:2410.02962) [10 citations]
- 129 Starkman, N. et al. (incl. **APW**), *Stream Members Only: Data-driven Characterization of Stellar Streams with Mixture Density Networks*, ApJ, 980, 253, 2025 (arXiv:2311.16960) [13 citations]
- 128 **Price-Whelan, A. M.**; Hunt, J. A. S.; Horta, D.; Oeur, M. et al., *Data-driven Dynamics with Orbital Torus Imaging: A Flexible Model of the Vertical Phase Space of the Galaxy*, ApJ, 979, 115, 2025 (arXiv:2401.07903) [9 citations]
- 127 Foote, H. R. et al. (incl. **APW**), *Segue 2 Recently Collided with the Cetus-Palca Stream: New Opportunities to Constrain Dark Matter in an Ultra-faint Dwarf*, ApJ, 979, 171, 2025 (arXiv:2408.06415) [3 citations]
- 126 Wagg, T.; Breivik, K.; Renzo, M.; **Price-Whelan, A. M.**, *cogsworth: A Gala of COSMIC proportions combining binary stellar evolution and galactic dynamics*, JOSS, 10, 7400, 2025 [5 citations]
- 125 Wagg, T.; Breivik, K.; Renzo, M.; **Price-Whelan, A. M.**, *cogsworth: A Gala of COSMIC Proportions Combining Binary Stellar Evolution and Galactic Dynamics*, ApJS, 276, 16, 2025 (arXiv:2409.04543) [13 citations]
- 124 Brooks, R. A. N.; Garavito-Camargo, N.; Johnston, K. V.; **Price-Whelan, A. M.** et al., *LMC Calls, Milky Way Halo Answers: Disentangling the Effects of the MW-LMC Interaction on Stellar Stream Populations*, ApJ, 978, 79, 2025 (arXiv:2410.02574) [20 citations]
- 123 Hackshaw, Z. et al. (incl. **APW**), *[X/Fe] Marks the Spot: Mapping Chemical Azimuthal Variations in the Galactic Disk with APOGEE*, ApJ, 977, 143, 2024 (arXiv:2405.18120) [25 citations]
- 122 Sagear, S.; **Price-Whelan, A. M.**; Ballard, S.; Lu, Y. (et al., *zoomies: A Tool to Infer Stellar Age*

- from Vertical Action in Gaia Data*, ApJ, 977, 49, 2024 (arXiv:2403.09878) [7 citations]
- 121 Garavito-Camargo, N.; **Price-Whelan, A. M.**; Samuel, J.; Cunningham, E. C. et al., *On the Corotation of Milky Way Satellites: LMC-mass Satellites Induce Apparent Motions in Outer Halo Tracers*, ApJ, 975, 100, 2024 (arXiv:2311.11359) [20 citations]
- 120 Foster, S. et al. (incl. **APW**), *Carbon enrichment in APOGEE disk stars as evidence of mass transfer in binaries*, A&A, 689, 2024 (arXiv:2407.18130) [6 citations]
- 119 Horta, D. et al. (incl. **APW**), *Stellar Mergers or Truly Young? Intermediate-age Stars on Highly Radial Orbits in the Milky Way's Stellar Halo*, ApJ, 971, 170, 2024 (arXiv:2403.09777) [19 citations]
- 118 Castro-Ginard, A. et al. (incl. **APW**), *Gaia DR3 detectability of unresolved binary systems*, A&A, 688, 2024 (arXiv:2404.14127) [102 citations]
- 117 Gaia Collaboration et al. (incl. **APW**), *Discovery of a dormant 33 solar-mass black hole in pre-release Gaia astrometry*, A&A, 686, 2024 (arXiv:2404.10486) [176 citations]
- 116 Cunningham, E. C.; Hunt, J. A. S.; **Price-Whelan, A. M.**; Johnston, K. V. et al., *Chemical Cartography of the Sagittarius Stream with Gaia*, ApJ, 963, 95, 2024 (arXiv:2307.08730) [30 citations]
- 115 Hunt, J. A. S.; **Price-Whelan, A. M.**; Johnston, K. V.; McClure, R. L. et al., *Radial phase spirals in the Solar neighbourhood*, MNRAS, 527, 11393, 2024 (arXiv:2401.08748) [17 citations]
- 114 Horta, D.; **Price-Whelan, A. M.**; Hogg, D. W.; Johnston, K. V. et al., *Orbital Torus Imaging: Acceleration, Density, and Dark Matter in the Galactic Disk Measured with Element Abundance Gradients*, ApJ, 962, 165, 2024 (arXiv:2312.07664) [22 citations]
- 113 Aganze, C. et al. (incl. **APW**), *Prospects for Detecting Gaps in Globular Cluster Stellar Streams in External Galaxies with the Nancy Grace Roman Space Telescope*, ApJ, 962, 151, 2024 (arXiv:2305.12045) [15 citations]
- 112 Darragh-Ford, E.; Hunt, J. A. S.; **Price-Whelan, A. M.**; Johnston, K. V., *ESCARGOT: Mapping Vertical Phase Spiral Characteristics Throughout the Real and Simulated Milky Way*, ApJ, 955, 74, 2023 (arXiv:2302.09086) [34 citations]
- 111 Yavetz, T. D.; Johnston, K. V.; Pearson, S.; **Price-Whelan, A. M.** et al., *Stream Fanning and Bifurcations: Observable Signatures of Resonances in Stellar Stream Morphology*, ApJ, 954, 215, 2023 (arXiv:2212.11006) [19 citations]
- 110 Castro-Ginard, A. et al. (incl. **APW**), *Estimating the selection function of Gaia DR3 subsamples*, A&A, 677, 2023 (arXiv:2303.17738) [44 citations]
- 109 Lucey, M. et al. (incl. **APW**), *Carbon-enhanced metal-poor star candidates from BP/RP spectra in Gaia DR3*, MNRAS, 523, 4049, 2023 (arXiv:2206.08299) [36 citations]
- 108 Silva, É. et al. (incl. **APW**), *Extratidal Members of Segue 3 are Rare and Difficult to Confirm*, RNAAS, 7, 127, 2023
- 107 Hawkins, K.; **Price-Whelan, A. M.**; Sheffield, A. A.; Subrahimovic, A. Z. et al., *On the Hunt for the Origins of the Orphan-Chenab Stream: Detailed Element Abundances with APOGEE and Gaia*, ApJ, 948, 123, 2023 (arXiv:2205.14218) [13 citations]
- 106 Lucey, M. et al. (incl. **APW**), *Dynamically constraining the length of the Milky way bar*, MNRAS, 520, 4779, 2023 (arXiv:2206.01798) [52 citations]

- 105 Cantat-Gaudin, T. et al. (incl. **APW**), *An empirical model of the Gaia DR3 selection function*, A&A, 669, 2023 (arXiv:2208.09335) [89 citations]
- 104 Chamberlain, K.; **Price-Whelan, A. M.**; Besla, G.; Cunningham, E. C. et al., *Implications of the Milky Way Travel Velocity for Dynamical Mass Estimates of the Local Group*, ApJ, 942, 18, 2023 (arXiv:2204.07173) [34 citations]
- 103 Rix, H.; Chandra, V.; Andrae, R.; **Price-Whelan, A. M.** et al., *The Poor Old Heart of the Milky Way*, ApJ, 941, 45, 2022 (arXiv:2209.02722) [141 citations]
- 102 Pearson, S.; **Price-Whelan, A. M.**; Hogg, D. W.; Seth, A. C. et al., *Mapping Dark Matter with Extragalactic Stellar Streams: The Case of Centaurus A*, ApJ, 941, 19, 2022 (arXiv:2205.12277) [40 citations]
- 101 Hunt, J. A. S.; **Price-Whelan, A. M.**; Johnston, K. V.; Darragh-Ford, E., *Multiple phase spirals suggest multiple origins in Gaia DR3*, MNRAS, 516, 2022 (arXiv:2206.06125) [68 citations]
- 100 Dillamore, A. M.; Belokurov, V.; Evans, N. W.; **Price-Whelan, A. M.**, *The impact of a massive Sagittarius dSph on GD-1-like streams*, MNRAS, 516, 1685, 2022 (arXiv:2205.13547) [32 citations]
- 99 Anguiano, B. et al. (incl. **APW**), *White Dwarf Binaries across the H-R Diagram*, AJ, 164, 126, 2022 (arXiv:2207.13992) [18 citations]
- 98 Yoshida, S.; Grunblatt, S.; **Price-Whelan, A. M.**, *Constraining the Planet Occurrence Rate around Halo Stars of Potentially Extragalactic Origin*, AJ, 164, 119, 2022 (arXiv:2206.13556) [6 citations]
- 97 Myers, N. et al. (incl. **APW**), *The Open Cluster Chemical Abundances and Mapping Survey. VI. Galactic Chemical Gradient Analysis from APOGEE DR17*, AJ, 164, 85, 2022 (arXiv:2206.13650) [77 citations]
- 96 Astropy Collaboration; **Price-Whelan, A. M.**; Lim, P. L.; Earl, N. et al., *The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package*, ApJ, 935, 167, 2022 (arXiv:2206.14220) [5700 citations]
- 95 Angus, R.; **Price-Whelan, A. M.**; Zinn, J. C.; Bedell, M. et al., *The 3D Galactocentric Velocities of Kepler Stars: Marginalizing Over Missing Radial Velocities*, AJ, 164, 25, 2022 (arXiv:2205.08901) [5 citations]
- 94 El-Badry, K. et al. (incl. **APW**), *Unicorns and giraffes in the binary zoo: stripped giants with subgiant companions*, MNRAS, 512, 5620, 2022 (arXiv:2203.06348) [72 citations]
- 93 Abdurro'uf et al. (incl. **APW**), *The Seventeenth Data Release of the Sloan Digital Sky Surveys: Complete Release of MaNGA, MaStar, and APOGEE-2 Data*, ApJS, 259, 35, 2022 (arXiv:2112.02026) [1448 citations]
- 92 Gandhi, S. S.; Johnston, K. V.; Hunt, J. A. S.; **Price-Whelan, A. M.** et al., *Snails across Scales: Local and Global Phase-mixing Structures as Probes of the Past and Future Milky Way*, ApJ, 928, 80, 2022 (arXiv:2107.03562) [28 citations]
- 91 Eilers, A. et al. (incl. **APW**), *Stellar Abundance Maps of the Milky Way Disk*, ApJ, 928, 23, 2022 (arXiv:2112.03295) [56 citations]
- 90 Hedges, C. et al. (incl. **APW**), *Erratum: "TOI-2076 and TOI-1807: Two Young, Comoving Planetary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up"*

- (2021, *AJ*, 162, 54), *AJ*, 163, 143, 2022 [3 citations]
- 89 Ness, M. K. et al. (incl. **APW**), *The Homogeneity of the Star-forming Environment of the Milky Way Disk over Time*, *ApJ*, 926, 144, 2022 (arXiv:2109.05722) [50 citations]
 - 88 Hedges, C. et al. (incl. **APW**), *Erratum: "TOI-2076 and TOI-1807: Two young, Comoving Planetary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up"* (2021, *AJ*, 162, 54), *AJ*, 162, 305, 2021
 - 87 Hasselquist, S. et al. (incl. **APW**), *APOGEE Chemical Abundance Patterns of the Massive Milky Way Satellites*, *ApJ*, 923, 172, 2021 (arXiv:2109.05130) [171 citations]
 - 86 Garavito-Camargo, N.; Patel, E.; Besla, G.; **Price-Whelan, A. M.** et al., *The Clustering of Orbital Poles Induced by the LMC: Hints for the Origin of Planes of Satellites*, *ApJ*, 923, 140, 2021 (arXiv:2108.07321) [40 citations]
 - 85 Santana, F. A. et al. (incl. **APW**), *Final Targeting Strategy for the SDSS-IV APOGEE-2S Survey*, *AJ*, 162, 303, 2021 (arXiv:2108.11908) [111 citations]
 - 84 Beaton, R. L. et al. (incl. **APW**), *Final Targeting Strategy for the Sloan Digital Sky Survey IV Apache Point Observatory Galactic Evolution Experiment 2 North Survey*, *AJ*, 162, 302, 2021 (arXiv:2108.11907) [111 citations]
 - 83 Kounkel, M.; Covey, K. R.; Stassun, K. G.; **Price-Whelan, A. M.** et al., *Double-lined Spectroscopic Binaries in the APOGEE DR16 and DR17 Data*, *AJ*, 162, 184, 2021 (arXiv:2107.10860) [86 citations]
 - 82 Garavito-Camargo, N.; Besla, G.; Laporte, C. F. P.; **Price-Whelan, A. M.** et al., *Quantifying the Impact of the Large Magellanic Cloud on the Structure of the Milky Way's Dark Matter Halo Using Basis Function Expansions*, *ApJ*, 919, 109, 2021 (arXiv:2010.00816) [136 citations]
 - 81 Rix, H. et al. (incl. **APW**), *Selection Functions in Astronomical Data Modeling, with the Space Density of White Dwarfs as a Worked Example*, *AJ*, 162, 142, 2021 (arXiv:2106.07653) [42 citations]
 - 80 Grunblatt, S. K.; Zinn, J. C.; **Price-Whelan, A. M.**; Angus, R. et al., *Age-dating Red Giant Stars Associated with Galactic Disk and Halo Substructures*, *ApJ*, 916, 88, 2021 (arXiv:2105.10505) [37 citations]
 - 79 Hedges, C. et al. (incl. **APW**), *TOI-2076 and TOI-1807: Two Young, Comoving Planetary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up*, *AJ*, 162, 54, 2021 (arXiv:2111.01311) [52 citations]
 - 78 Foreman-Mackey, D. et al. (incl. **APW**), *exoplanet: Gradient-based probabilistic inference for exoplanet data & other astronomical time series*, *JOSS*, 6, 3285, 2021 (arXiv:2105.01994) [233 citations]
 - 77 Putman, M. E.; Zheng, Y.; **Price-Whelan, A. M.**; Grcevich, J. et al., *The Gas Content and Stripping of Local Group Dwarf Galaxies*, *ApJ*, 913, 53, 2021 (arXiv:2101.07809) [145 citations]
 - 76 Sheffield, A. A. et al. (incl. **APW**), *Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2*, *ApJ*, 913, 39, 2021 (arXiv:2103.07488) [5 citations]
 - 75 Valluri, M.; **Price-Whelan, A. M.**; Snyder, S. J., *Detecting the Figure Rotation of Dark Matter Halos with Tidal Streams*, *ApJ*, 910, 150, 2021 (arXiv:2009.09004) [20 citations]
 - 74 Miller, A. et al. (incl. **APW**), *Orbital and Stellar Parameters for 2M06464003+0109157:*

- A Double-lined Eclipsing Binary of Spotted, Sub-solar Twins*, PASP, 133, 44201, 2021 (arXiv:2103.10488) [4 citations]
- 73 **Price-Whelan, A. M.**; Hogg, D. W.; Johnston, K. V.; Ness, M. K. et al., *Orbital Torus Imaging: Using Element Abundances to Map Orbits and Mass in the Milky Way*, ApJ, 910, 17, 2021 (arXiv:2012.00015) [30 citations]
- 72 Yavetz, T. D.; Johnston, K. V.; Pearson, S.; **Price-Whelan, A. M.** et al., *Separatrix divergence of stellar streams in galactic potentials*, MNRAS, 501, 1791, 2021 (arXiv:2011.11919) [28 citations]
- 71 Mazzola, C. N. et al. (incl. **APW**), *The close binary fraction as a function of stellar parameters in APOGEE: a strong anticorrelation with α abundances*, MNRAS, 499, 1607, 2020 (arXiv:2007.09059) [60 citations]
- 70 Shipp, N.; **Price-Whelan, A. M.**; Tavangar, K.; Mateu, C. et al., *Discovery of Extended Tidal Tails around the Globular Cluster Palomar 13*, AJ, 160, 244, 2020 (arXiv:2006.12501) [28 citations]
- 69 Angus, R.; Beane, A.; **Price-Whelan, A. M.**; Newton, E. et al., *Exploring the Evolution of Stellar Rotation Using Galactic Kinematics*, AJ, 160, 90, 2020 (arXiv:2005.09387) [52 citations]
- 68 Ahumada, R. et al. (incl. **APW**), *The 16th Data Release of the Sloan Digital Sky Surveys: First Release from the APOGEE-2 Southern Survey and Full Release of eBOSS Spectra*, ApJS, 249, 3, 2020 (arXiv:1912.02905) [1486 citations]
- 67 Caldwell, N.; Bonaca, A.; **Price-Whelan, A. M.**; Sesar, B. et al., *A Larger Extent for the Ophiuchus Stream*, AJ, 159, 287, 2020 (arXiv:2004.14350) [15 citations]
- 66 **Price-Whelan, A. M.**; Hogg, D. W.; Rix, H.; Beaton, R. L. et al., *Close Binary Companions to APOGEE DR16 Stars: 20,000 Binary-star Systems Across the Color-Magnitude Diagram*, ApJ, 895, 2, 2020 (arXiv:2002.00014) [130 citations]
- 65 Mumford, S. et al. (incl. **APW**), *SunPy: A Python package for Solar Physics*, JOSS, 5, 1832, 2020 [49 citations]
- 64 Hayes, C. R. et al. (incl. **APW**), *Metallicity and α -Element Abundance Gradients along the Sagittarius Stream as Seen by APOGEE*, ApJ, 889, 63, 2020 (arXiv:1912.06707) [83 citations]
- 63 Bonaca, A.; Pearson, S.; **Price-Whelan, A. M.**; Dey, A. et al., *Variations in the Width, Density, and Direction of the Palomar 5 Tidal Tails*, ApJ, 889, 70, 2020 (arXiv:1910.00592) [76 citations]
- 62 Nidever, D. L.; **Price-Whelan, A. M.**; Choi, Y.; Beaton, R. L. et al., *Spectroscopy of the Young Stellar Association Price-Whelan 1: Origin in the Magellanic Leading Arm and Constraints on the Milky Way Hot Halo*, ApJ, 887, 115, 2019 (arXiv:1910.05360) [24 citations]
- 61 **Price-Whelan, A. M.**; Nidever, D. L.; Choi, Y.; Schlafly, E. F. et al., *Discovery of a Disrupting Open Cluster Far into the Milky Way Halo: A Recent Star Formation Event in the Leading Arm of the Magellanic Stream?*, ApJ, 887, 19, 2019 (arXiv:1811.05991) [38 citations]
- 60 **Price-Whelan, A. M.**; Mateu, C.; Iorio, G.; Pearson, S. et al., *Kinematics of the Palomar 5 Stellar Stream from RR Lyrae Stars*, AJ, 158, 223, 2019 (arXiv:1910.00595) [43 citations]
- 59 Chakrabarti, S.; Chang, P.; **Price-Whelan, A. M.**; Read, J. et al., *Antlia 2's Role in Driving the Ripples in the Outer Gas Disk of the Galaxy*, ApJ, 886, 67, 2019 (arXiv:1906.04203) [21 ci-

tations]

- 58 Koppelman, H. H.; Helmi, A.; Massari, D.; **Price-Whelan, A. M.** et al., [Multiple retrograde substructures in the Galactic halo: A shattered view of Galactic history](#), A&A, 631, 2019 (arXiv:1909.08924) [261 citations]
- 57 Kollmeier, J. et al. (incl. **APW**), [SDSS-V Pioneering Panoptic Spectroscopy](#), BAAS, 51, 274, 2019 [22 citations]
- 56 Norman, D. et al. (incl. **APW**), [The Growing Importance of a Tech Savvy Astronomy and Astrophysics Workforce](#), BAAS, 51, 18, 2019 (arXiv:1910.08376) [2 citations]
- 55 Tollerud, E.; Smith, A.; **Price-Whelan, A. M.**; Cruz, K. et al., [Sustaining Community-Driven Software for Astronomy in the 2020s](#), BAAS, 51, 180, 2019
- 54 Bauer, A. et al. (incl. **APW**), [A Need for Dedicated Outreach Expertise and Online Programming](#), BAAS, 51, 130, 2019 (arXiv:1910.14088)
- 53 Smith, A. et al. (incl. **APW**), [Elevating the Role of Software as a Product of the Research Enterprise](#), BAAS, 51, 52, 2019 (arXiv:1907.06981)
- 52 Casey, A. R. et al. (incl. **APW**), [Tidal Interactions between Binary Stars Can Drive Lithium Production in Low-mass Red Giants](#), ApJ, 880, 125, 2019 (arXiv:1902.04102) [104 citations]
- 51 Erkal, D. et al. (incl. **APW**), [The total mass of the Large Magellanic Cloud from its perturbation on the Orphan stream](#), MNRAS, 487, 2685, 2019 (arXiv:1812.08192) [377 citations]
- 50 Bonaca, A.; Hogg, D. W.; **Price-Whelan, A. M.**; Conroy, C., [The Spur and the Gap in GD-1: Dynamical Evidence for a Dark Substructure in the Milky Way Halo](#), ApJ, 880, 38, 2019 (arXiv:1811.03631) [257 citations]
- 49 Rasskazov, A. et al. (incl. **APW**), [Hypervelocity Stars from a Supermassive Black Hole-Intermediate-mass Black Hole Binary](#), ApJ, 878, 17, 2019 (arXiv:1810.12354) [37 citations]
- 48 Koposov, S. E. et al. (incl. **APW**), [Piercing the Milky Way: an all-sky view of the Orphan Stream](#), MNRAS, 485, 4726, 2019 (arXiv:1812.08172) [140 citations]
- 47 Williams, B. et al. (incl. **APW**), [Far Reaching Science with Resolved Stellar Populations in the 2020s](#), BAAS, 51, 301, 2019
- 46 **Price-Whelan, A. M.**; Breivik, K.; D'Orazio, D.; Hogg, D. W. et al., [Stellar multiplicity: an interdisciplinary nexus](#), BAAS, 51, 206, 2019 [3 citations]
- 45 Sanderson, R. et al. (incl. **APW**), [The Multidimensional Milky Way](#), BAAS, 51, 347, 2019 (arXiv:1903.07641) [8 citations]
- 44 Ness, M. et al. (incl. **APW**), [In Pursuit of Galactic Archaeology](#), BAAS, 51, 238, 2019
- 43 Li, T. et al. (incl. **APW**), [Dark Matter Physics with Wide Field Spectroscopic Surveys](#), BAAS, 51, 252, 2019
- 42 **Price-Whelan, A. M.**; Goodman, J., [Binary Companions of Evolved Stars in APOGEE DR14: Orbital Circularization](#), ApJ, 867, 5, 2018 (arXiv:1804.06841) [35 citations]
- 41 De Rosa, G. et al. (incl. **APW**), [Velocity-resolved Reverberation Mapping of Five Bright Seyfert 1 Galaxies](#), ApJ, 866, 133, 2018 (arXiv:1807.04784) [104 citations]
- 40 Kado-Fong, E.; Greene, J. E.; Hendel, D.; **Price-Whelan, A. M.** et al., [Tidal Features at 0.05 z 0.45 in the Hyper Suprime-Cam Subaru Strategic Program: Properties and Formation Channels](#), ApJ, 866, 103, 2018 (arXiv:1805.05970) [72 citations]

- 39 Anderson, L.; Hogg, D. W.; Leistedt, B.; **Price-Whelan, A. M.** et al., *Improving Gaia Parallax Precision with a Data-driven Model of Stars*, AJ, 156, 145, 2018 (arXiv:1706.05055) [31 citations]
- 38 Astropy Collaboration; **Price-Whelan, A. M.**; Sipócz, B. M.; Günther, H. M. et al., *The Astropy Project: Building an Open-science Project and Status of the v2.0 Core Package*, AJ, 156, 123, 2018 (arXiv:1801.02634) [10891 citations]
- 37 Hendel, D. et al. (incl. **APW**), *SMHASH: anatomy of the Orphan Stream using RR Lyrae stars*, MNRAS, 479, 570, 2018 (arXiv:1711.04663) [22 citations]
- 36 **Price-Whelan, A. M.**; Hogg, D. W.; Rix, H.; De Lee, N. et al., *Binary Companions of Evolved Stars in APOGEE DR14: Search Method and Catalog of ~5000 Companions*, AJ, 156, 18, 2018 (arXiv:1804.04662) [1326 citations]
- 35 Bergemann, M. et al. (incl. **APW**), *Two chemically similar stellar overdensities on opposite sides of the plane of the Galactic disk*, Nature, 555, 334, 2018 (arXiv:1803.00563) [66 citations]
- 34 Morris, B. M. et al. (incl. **APW**), *astrophys: An Open Source Observation Planning Package in Python*, AJ, 155, 128, 2018 (arXiv:1712.09631) [85 citations]
- 33 Oh, S.; **Price-Whelan, A. M.**; Brewer, J. M.; Hogg, D. W. et al., *Kronos and Krios: Evidence for Accretion of a Massive, Rocky Planetary System in a Comoving Pair of Solar-type Stars*, ApJ, 854, 138, 2018 (arXiv:1709.05344) [100 citations]
- 32 Sheffield, A. A.; **Price-Whelan, A. M.**; Tzanidakis, A.; Johnston, K. V. et al., *A Disk Origin for the Monoceros Ring and A13 Stellar Overdensities*, ApJ, 854, 47, 2018 (arXiv:1801.01171) [38 citations]
- 31 Greco, J. P.; Greene, J. E.; **Price-Whelan, A. M.**; Leauthaud, A. et al., *Sumo Puff: Tidal debris or disturbed ultra-diffuse galaxy?*, PASJ, 70, 2018 (arXiv:1704.06681) [22 citations]
- 30 Goulding, A. D. et al. (incl. **APW**), *Galaxy interactions trigger rapid black hole growth: An unprecedented view from the Hyper Suprime-Cam survey*, PASJ, 70, 2018 (arXiv:1706.07436) [202 citations]
- 29 **Price-Whelan, A. M.**, *Gala: A Python package for galactic dynamics*, JOSS, 2, 388, 2017 [360 citations]
- 28 Alam, S. et al. (incl. **APW**), *The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey: cosmological analysis of the DR12 galaxy sample*, MNRAS, 470, 2617, 2017 (arXiv:1607.03155) [3109 citations]
- 27 **Price-Whelan, A. M.**; Foreman-Mackey, D., *schwimmbad: A uniform interface to parallel processing pools in Python*, JOSS, 2, 357, 2017 [43 citations]
- 26 Pearson, S.; **Price-Whelan, A. M.**; Johnston, K. V., *Gaps and length asymmetry in the stellar stream Palomar 5 as effects of Galactic bar rotation*, Nature Astronomy, 1, 633, 2017 (arXiv:1703.04627) [138 citations]
- 25 Johnston, K. V.; **Price-Whelan, A. M.**; Bergemann, M.; Laporte, C. et al., *Disk Heating, Galactoseismology, and the Formation of Stellar Halos*, MDPI: galaxies, 5, 44, 2017 (arXiv:1709.00491) [12 citations]
- 24 Li, T. S. et al. (incl. **APW**), *Exploring Halo Substructure with Giant Stars. XV. Discovery of a Connection between the Monoceros Ring and the Triangulum-Andromeda Overdensity?*, ApJ,

- 844, 74, 2017 ([arXiv:1703.05384](#)) [37 citations]
- 23 [Oh, S.](#); **Price-Whelan, A. M.**; Hogg, D. W.; Morton, T. D. et al., *Comoving Stars in Gaia DR1: An Abundance of Very Wide Separation Comoving Pairs*, *AJ*, 153, 257, 2017 ([arXiv:1612.02440](#)) [171 citations]
 - 22 [Sesar, B.](#); [Fouesneau, M.](#); **Price-Whelan, A. M.**; [Bailer-Jones, C. A. L.](#) et al., *A Probabilistic Approach to Fitting Period-luminosity Relations and Validating Gaia Parallaxes*, *ApJ*, 838, 107, 2017 ([arXiv:1611.07035](#)) [51 citations]
 - 21 **Price-Whelan, A. M.**; Hogg, D. W.; Foreman-Mackey, D.; Rix, H., *The Joker: A Custom Monte Carlo Sampler for Binary-star and Exoplanet Radial Velocity Data*, *ApJ*, 837, 20, 2017 ([arXiv:1610.07602](#)) [146 citations]
 - 20 [Charisi, M.](#); [Bartos, I.](#); [Haiman, Z.](#); **Price-Whelan, A. M.** et al., *A population of short-period variable quasars from PTF as supermassive black hole binary candidates*, *MNRAS*, 463, 2145, 2016 ([arXiv:1604.01020](#)) [269 citations]
 - 19 **Price-Whelan, A. M.**; [Sesar, B.](#); [Johnston, K. V.](#); Rix, H., *Spending Too Much Time at the Galactic Bar: Chaotic Fanning of the Ophiuchus Stream*, *ApJ*, 824, 104, 2016 ([arXiv:1601.06790](#)) [63 citations]
 - 18 **Price-Whelan, A. M.**; [Johnston, K. V.](#); [Valluri, M.](#); [Pearson, S.](#) et al., *Chaotic dispersal of tidal debris*, *MNRAS*, 455, 1079, 2016 ([arXiv:1507.08662](#)) [74 citations]
 - 17 [Charisi, M.](#); [Bartos, I.](#); [Haiman, Z.](#); **Price-Whelan, A. M.** et al., *Multiple periods in the variability of the supermassive black hole binary candidate quasar PG1302-102?*, *MNRAS*, 454, 2015 ([arXiv:1502.03113](#)) [29 citations]
 - 16 **Price-Whelan, A. M.**; [Johnston, K. V.](#); [Sheffield, A. A.](#); [Laporte, C. F. P.](#) et al., *A reinterpretation of the Triangulum-Andromeda stellar clouds: a population of halo stars kicked out of the Galactic disc*, *MNRAS*, 452, 676, 2015 ([arXiv:1503.08780](#)) [99 citations]
 - 15 [Sesar, B.](#) et al. (incl. **APW**), *The Nature and Orbit of the Ophiuchus Stream*, *ApJ*, 809, 59, 2015 ([arXiv:1501.00581](#)) [32 citations]
 - 14 [Alam, S.](#) et al. (incl. **APW**), *The Eleventh and Twelfth Data Releases of the Sloan Digital Sky Survey: Final Data from SDSS-III*, *ApJS*, 219, 12, 2015 ([arXiv:1501.00963](#)) [2417 citations]
 - 13 [Pearson, S.](#); [Küpper, A. H. W.](#); [Johnston, K. V.](#); **Price-Whelan, A. M.**, *Tidal Stream Morphology as an Indicator of Dark Matter Halo Geometry: The Case of Palomar 5*, *ApJ*, 799, 28, 2015 ([arXiv:1410.3477](#)) [83 citations]
 - 12 **Price-Whelan, A. M.**; Hogg, D. W.; [Johnston, K. V.](#); [Hendel, D.](#), *Inferring the Gravitational Potential of the Milky Way with a Few Precisely Measured Stars*, *ApJ*, 794, 4, 2014 ([arXiv:1405.6721](#)) [58 citations]
 - 11 [Anderson, L.](#) et al. (incl. **APW**), *The clustering of galaxies in the SDSS-III Baryon Oscillation Spectroscopic Survey: baryon acoustic oscillations in the Data Releases 10 and 11 Galaxy samples*, *MNRAS*, 441, 24, 2014 ([arXiv:1312.4877](#)) [1501 citations]
 - 10 [Ahn, C. P.](#) et al. (incl. **APW**), *The Tenth Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-III Apache Point Observatory Galactic Evolution Experiment*, *ApJS*, 211, 17, 2014 ([arXiv:1307.7735](#)) [954 citations]
 - 9 **Price-Whelan, A. M.**; [Agüeros, M. A.](#); [Fournier, A. P.](#); [Street, R.](#) et al., *Statistical Searches*

- for Microlensing Events in Large, Non-uniformly Sampled Time-Domain Surveys: A Test Using Palomar Transient Factory Data*, ApJ, 781, 35, 2014 (arXiv:1311.3683) [14 citations]
- 8 Astropy Collaboration et al. (incl. **APW**), *Astropy: A community Python package for astronomy*, A&A, 558, 2013 (arXiv:1307.6212) [14485 citations]
 - 7 Dawson, K. S. et al. (incl. **APW**), *The Baryon Oscillation Spectroscopic Survey of SDSS-III*, AJ, 145, 10, 2013 (arXiv:1208.0022) [2085 citations]
 - 6 Ahn, C. P. et al. (incl. **APW**), *The Ninth Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-III Baryon Oscillation Spectroscopic Survey*, ApJS, 203, 21, 2012 (arXiv:1207.7137) [1370 citations]
 - 5 Eisenstein, D. J. et al. (incl. **APW**), *SDSS-III: Massive Spectroscopic Surveys of the Distant Universe, the Milky Way, and Extra-Solar Planetary Systems*, AJ, 142, 72, 2011 (arXiv:1101.1529) [2061 citations]
 - 4 Aihara, H. et al. (incl. **APW**), *Erratum: "The Eighth Data Release of the Sloan Digital Sky Survey: First Data from SDSS-III"* (2011, ApJS, 193, 29), ApJS, 195, 26, 2011 [59 citations]
 - 3 Blanton, M. R. et al. (incl. **APW**), *Improved Background Subtraction for the Sloan Digital Sky Survey Images*, AJ, 142, 31, 2011 (arXiv:1105.1960) [552 citations]
 - 2 Aihara, H. et al. (incl. **APW**), *The Eighth Data Release of the Sloan Digital Sky Survey: First Data from SDSS-III*, ApJS, 193, 29, 2011 (arXiv:1101.1559) [1361 citations]
 - 1 **Price-Whelan, A. M.**; Hogg, D. W., *What Bandwidth Do I Need for My Image?*, PASP, 122, 207, 2010 (arXiv:0910.2375) [4 citations]

Preprints & other

- 23 Pearson, S.; Nibauer, J.; Cunningham, E. C.; **Price-Whelan, A. M.** et al., *Extragalactic Stellar Streams in Time-Dependent Cosmological Halos*, 2026 (arXiv:2609.00526)
- 22 Dubay, L. O. et al. (incl. **APW**), *A Broken Clock Is Right Twice a Day: [Ce/Mg] Is Not a Universal Chemical Clock*, 2026 (arXiv:2608.26261)
- 21 Kiel Holm, J. et al. (incl. **APW**), *Evidence for the First Globular Cluster Stellar Stream beyond the Milky Way*, 2026 (arXiv:2608.12254)
- 20 SDSS Collaboration et al. (incl. **APW**), *The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper*, 2026 (arXiv:2607.26149)
- 19 Wagg, T.; Breivik, K.; **Price-Whelan, A. M.**; Renzo, M. et al., *Charting the Galactic Underworld I: Comprehensive simulations of the kinematics, rates, and demographics of Milky Way black holes*, 2026 (arXiv:2607.22814)
- 18 Han, J. J. et al. (incl. **APW**), *A Path to an All-Sky Survey with Roman*, 2026 (arXiv:2602.21280)
- 17 Sayeed, M. et al. (incl. **APW**), *Looking for Companionship: Radial Velocity Follow-Up of Lithium-Rich Giants with ESPRESSO*, 2025 (arXiv:2510.17966)
- 16 Sagar, S.; Ballard, S.; Daniel, K. J.; **Price-Whelan, A. M.** et al., *The Orbital Eccentricities of Planets in the Kinematic Thin and Thick Galactic Disks*, 2025 (arXiv:2509.23973) [3 citations]

- 15 Blanton, M. R. et al. (incl. **APW**), *The Future of Astronomical Data Infrastructure: Meeting Report*, 2023 (arXiv:2311.04272)
- 14 Dey, A. et al. (incl. **APW**), *RomAndromeda: The Roman Survey of the Andromeda Halo*, 2023 (arXiv:2306.12302) [6 citations]
- 13 Han, J. J.; Dey, A.; **Price-Whelan, A. M.**; Najita, J. et al., *NANCY: Next-generation All-sky Near-infrared Community survey*, 2023 (arXiv:2306.11784) [10 citations]
- 12 Grunblatt, S. K. et al. (incl. **APW**), *Roman CCS White Paper: Adding Fields Hosting Globular Clusters To The Galactic Bulge Time Domain Survey*, 2023 (arXiv:2306.10647) [5 citations]
- 11 Breivik, K. et al. (incl. **APW**), *From Data to Software to Science with the Rubin Observatory LSST*, 2022 (arXiv:2208.02781) [17 citations]
- 10 Bechtol, K. et al. (incl. **APW**), *Snowmass2021 Cosmic Frontier White Paper: Dark Matter Physics from Halo Measurements*, 2022 (arXiv:2203.07354) [70 citations]
- 9 Katz, D. S. et al. (incl. **APW**), *Software Sustainability & High Energy Physics*, 2020 (arXiv:2010.05102) [2 citations]
- 8 Oladosu, A. et al. (incl. **APW**), *Meta-Learning for One-Class Classification with Few Examples using Order-Equivariant Network*, 2020 (arXiv:2007.04459) [4 citations]
- 7 Hogg, D. W.; **Price-Whelan, A. M.**; Leistedt, B., *Data Analysis Recipes: Products of multivariate Gaussians in Bayesian inferences*, 2020 (arXiv:2005.14199) [14 citations]
- 6 Ness, M. et al. (incl. **APW**), *In Pursuit of Galactic Archaeology: Astro2020 Science White Paper*, 2019 (arXiv:1907.05422)
- 5 Buckley, M. R.; Hogg, D. W.; **Price-Whelan, A. M.**, *Applying Liouville's Theorem to Gaia Data*, 2019 (arXiv:1907.00987) [3 citations]
- 4 The MSE Science Team et al. (incl. **APW**), *The Detailed Science Case for the Maunakea Spectroscopic Explorer, 2019 edition*, 2019 (arXiv:1904.04907) [89 citations]
- 3 Breivik, K.; **Price-Whelan, A. M.**; D'Orazio, D. J.; Hogg, D. W. et al., *Stellar multiplicity: an interdisciplinary nexus*, 2019 (arXiv:1903.05094) [4 citations]
- 2 Bergemann, M. et al. (incl. **APW**), *Stellar Astrophysics and Exoplanet Science with the Maunakea Spectroscopic Explorer (MSE)*, 2019 (arXiv:1903.03157) [2 citations]
- 1 **Price-Whelan, A. M.**; Oh, S.; Spergel, D. N., *Spectroscopic confirmation of very-wide stellar binaries and large-separation comoving pairs from Gaia DR1*, 2017 (arXiv:1709.03532) [18 citations]